MTH1004

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

May 2024

Probability, Statistics, and Data

Module Leader: Dr Stefan Siegert

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Statistical tables (Murdoch & Barnes) will be provided. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) (i) State the definition of the Binomial coefficient $\binom{n}{k}$ in terms of factorials.

   (ii) Find the number of ways to form a committee of 4 people from a group of 10 people (unordered sampling without replacement).

   (iii) State the Binomial theorem.

   (iv) Show that $\sum_{k=0}^{n} \binom{n}{k} = 2^n$. (10)

   (b) Consider two discrete random variables X and Y, both with range space $\{1, 2, 3\}$ with joint probability mass function (pmf) tabulated as follows:

$p_{X,Y}(x, y)$	$y = 1$	$y = 2$	$y = 3$
$x = 1$	0.15	0.15	0.1
$x = 2$	0.0	0.2	0.2
$x = 3$	0.15	0.05	0.0

   (i) Find $P(X = 1, Y = 3)$.

   (ii) Define the marginal pmf of X in terms of $p_{X,Y}(x, y)$, and calculate $P(X = 1)$.

   (iii) Define the conditional pmf of X given Y in terms of $p_{X,Y}(x, y)$, and calculate $P(X = 1|Y = 3)$. (10)

   (c) A study sought to investigate how the time taken for people to fall asleep is affected by having a warm bath before going to bed. The study split a simple random sample of 100 people into two groups of 50 people each. Those in group A took a bath before bed and those in group B did not. The time taken for each person to fall asleep was recorded. For group A, the sample mean time was $\bar{y}_A = 10$ minutes and the sample standard deviation was $s_A = 11$ minutes. For group B, the sample mean time was $\bar{y}_B = 18$ minutes and the sample standard deviation was $s_B = 20$ minutes.

   (i) Conduct a test of size 5% of the null hypothesis that the population mean times are the same for both groups against the alternative hypothesis that the population mean times differ. State clearly your test statistic, its null distribution, the critical region and your conclusion.

   (ii) Compute an equal-tailed 95% confidence interval for the difference in population mean times for the two groups. (15)

(d) (i) State the Central Limit Theorem.

(ii) Suppose that the number of goals scored by a football team in each match follows a Poisson distribution with mean and variance both equal to 2, and suppose that the numbers of goals scored are independent between matches. Use the Central Limit Theorem to calculate an approximate probability that the team scores at least 100 goals in a season of 38 matches. (15)

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SECTION B

2. Let A and B be events with $0 < P(A) < 1$ and $0 < P(B) < 1$.

   (a) Show that $P(A|B) = 1$ implies $P(B^c|A^c) = 1$. (6)

   (b) Show that $P(A|B) = 1$ implies $P(B|A) > P(B)$. (4)

   (c) Show that $P(A^c \cap (A \cup B)) = P(A^c \cap B)$ and hence, or otherwise, show that if A and B are independent, then A^c and B are independent, too. (15)

   [25]

3. Let $Y_1, \ldots, Y_n$ be independent random variables, each with probability density function

$$f(y) = \frac{2y}{\theta^2} \qquad \text{for } 0 \leq y \leq \theta,$$

   where θ is a parameter.

   (a) Show that the method of moments estimator for θ is $\hat{\theta} = 3\bar{Y}/2$, where $\bar{Y} = \sum_{i=1}^{n} Y_i/n$. (5)

   (b) Given that $\mathrm{E}(Y_1^2) = \theta^2/2$ (which you may use without proving it), calculate the bias and variance of $\hat{\theta}$. (10)

   (c) Consider the estimator $\tilde{\theta} = c\hat{\theta}$, where c is a constant. Find the value of c which minimises the mean squared error of $\tilde{\theta}$. (10)

   [25]

4. Consider the continuous random variable X with piecewise linear probability density function given by

$$f(x) = \begin{cases} 0 & \text{for } -\infty < x < a, \\ \frac{2}{a(a-b)}(x-a) & \text{for } a \leq x < 0, \\ \frac{2}{b(b-a)}(b-x) & \text{for } 0 \leq x < b, \\ 0 & \text{for } b \leq x < \infty, \end{cases}$$

where $a < 0$ and $b > 0$ are constant parameters. A sketch of $f(x)$ is given in Figure 1.

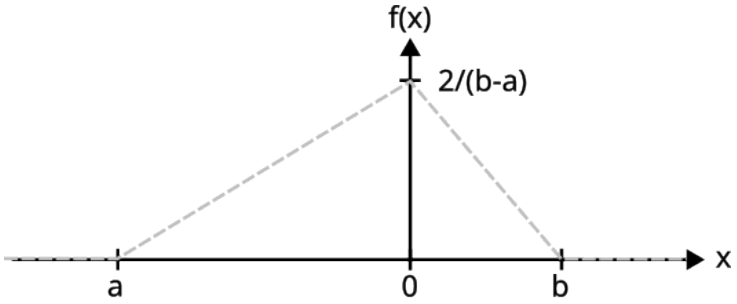


Figure 1: Sketch of $f(x)$ used in question 4.

(a) Show that the k-th moment of X is given by

$$\frac{2(b^{k+1} - a^{k+1})}{(b-a)(k+1)(k+2)}.$$

(Hint: Use direct calculation rather than the moment-generating function.) (12)

(b) Hence, or otherwise, calculate the expectation and variance of X. If possible, use the identities $b^2 - a^2 = (b+a)(b-a)$ and $b^3 - a^3 = (b-a)(b^2 + ab + a^2)$ to simplify your result. (9)

(c) Consider the random variable $Y = X + c$ where c is a finite real-valued constant. State the expectation and variance of Y, and briefly explain your reasoning. (4)

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